Data Processing Tools
Part 2

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Outline

- Data Processing
- Pandas

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Scientific Python Ecosystem

- **Core numeric libraries**
  - Numpy: Numerical computing
  - Scipy: High-level numerical functions, e.g., optimization, regression
  - Matplotlib: 2D visualization
  - Sympy: Symbolic computing
- **Domain-specific packages**
  - Pandas, Statmodels, Seaborn: Statistics
  - Scikit-learn: Machine learning
  - Mayavi: 3D visualization
  - Scikit-image: Image processing

Image Source: https://scipy-lectures.org/intro/intro.html

Scientific Python Ecosystem (cont.)

- Library hierarchies

Core

Domain-Specific

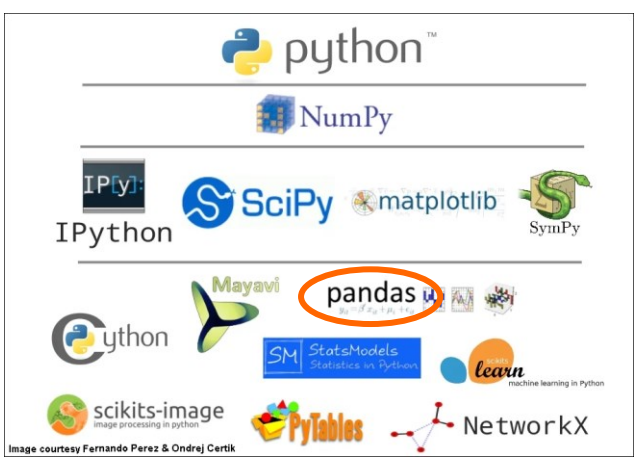

Image Source: https://speakerdeck.com/cournape/a-journey-through-the-scientific-python-ecosystem

Python Libraries for Data Science

- Top 10 Python libraries for Data Science

1.Tensorflow	1. NumPy	1. Pandas	1. NumPy
2. NumPy	2. Keras	2. NumPy	2. SciPy
3. SciPy	3. Pandas	3. SciPy	3. BeautifulSoup
4. Pandas	4. PyTorch	4. SciKit-Learn	4. Scrappy
5. Matplotlib	5. SciPy	5. Tensorflow	5. Tensorflow
6. Keras	6. SciKit-Learn	6. Keras	6. Pandas
7. SciKit-Learn	7. Tensorflow	7. Matplotlib	7. Matplotlib
8. PyTorch	8. Matplotlib	8. Seaborn	8. Plotly
9. Scrappy	9. Seaborn	9. Plotly	9. SciKit-Learn
10. BeautifulSoup	10. Theano	10. GGplot	10. PyCaret

Sources: https://www.simplilearn.com/top-python-libraries-for-data-science-article
https://builtin.com/data-science/python-libraries-data-science
https://www.geeksforgeeks.org/top-10-python-libraries-for-data-science-in-2021/
https://www.analyticsvidhya.com/blog/2020/11/top-13-python-libraries-every-data-science-aspirant-must-know-and-their-resources/
Image Source: https://makemeanalyst.com/data-science-with-python/python-libraries-for-data-analysis

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Pandas

- Shorten from "Panel Data"
- Python library for data manipulation and analysis
- Examples
  - Series and **Data Frame** data structures
  - In-memory data reading and writing
  - Data selection such as slicing, indexing
  - Data reshape such as merge, join

Image Source: https://en.wikipedia.org/wiki/Pandas_%28software%29#References

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Data Frame Motivation

- For real-world data, data are given in table format.
- To handle table-like data is crucial

Cartridge	Product Number	Bullet Make	Bullet Type	Bullet Weight Gr	COAL (Inches)
243 Winchester	M6000	Berger	VLD Hunting	90	2.775
6 Creedmoor		Berger	Hybrid Target	105	2.670
6XC	M6001	Berger	VLD Hunting	115	2.745
25-06 Rem	M6002	Berger	VLD Hunting	115	3.215
6.5 Creedmoor	M6004	Berger	Elite Hunter	140	2.865
6.5-284	M6006	Berger	VLD Hunting	140	3.085
264 Win Mag	M6007	Berger	VLD Hunting	140	3.325
6.5 PRC		Berger	Elite Hunter	140	2.970
280 Remington	M6012	Berger	VLD Hunting	140	3.330
280 Ackley	M6013	Berger	VLD Hunting	168	3.400
7mm RSAUM		Berger	VLD Hunting	168	2.800
7mm RSAUM	M6042	Berger	Hybrid Target	180	2.820
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7mm STW	M6018	Berger	VLD Hunting	180	3.670
				180	3.436
				180	3.665
				180	3.610
				195	3.630
				168	2.800
				168	3.380
				190	3.510
				190	3.650
				210	3.650
				190	3.650
				190	3.650
				215	3.555
				190	3.650
				230	3.650
				230	3.615
				300	3.550
				300	3.600

7mm Valkyrie Data sheet

Specifications

Cartrige over all length 2.295 inches

trim to length 1.596 max case length 1.605

Bullet	Bullet Weight	Ballistic Coefficent (BC)	Loaded COAL (In)	Powder	Powder Weight (Grains)	Primer	Pressure (psi)	Parent Brass	Average Velocity (FPS)	Group Size (In)
Standard disclaimer, use loads at own risk they were safe in the guns tested in. follow standard loading saftey										
110 grain										
Speer TNT	110	0.338	2.295	imr8208xbr	38.5	rem 7 1/2		lapua 6.5x47	3095	0.598
Speer TNT	110	0.338	2.285	imr8208xbr	37.9	cci 400		lapua 6.5x47	2843	0.5
Speer TNT	110	0.338	2.285	imr8208xbr	37.9	cci 400		lapua 6.5x47	2915	0.5
120 grain										
Hornady vmax	120	0.365	2.34	AA2230	34.2	rem 7 1/2		lapua 6.5x47	2850	0.4
Sierra Pro hunter	120	0.326	2.292	benchmark	36.5	rem 7 1/2		lapua 6.5x47	2970	0.65
Sierra Pro hunter	120	0.326	2.28	imr8208xbr	37.7	cci400		lapua 6.5x47		< .75
130 grain										
Speer sptt	130	0.424	2.26	Leverevolution	40.3	cci450		lapua 6.5x47	2704	< 1"
Speer sptt	130	0.424	2.26	Leverevolution	40.3	cci450		lapua 6.5x47	2799	< 1"
Speer sptt	130	0.424	2.26	CFE	40.3	cci450		lapua 6.5x47	2670	< 1"
Speer sptt	130	0.424	2.26	CFE	40.3	cci450		lapua 6.5x47	2779	< 1"
Speer sptt	130	0.424	2.26	CFE	40.5	rem 7 1/2		lapua 6.5x47	2986	0.68
139 grain										
Hornady sst	139	0.453	2.295	Leverevolution	38.2	rem 7 1/2		lapua 6.5x47	2775	0.925
Hornadey GMX	139	0.486	2.345	Leverevolution	38.2	CCI 41		lapua 6.5x47	2715	0.7

Geo data table

Edit row

Enter filter text

	Location	Geging station	River	Level	Date	Average maxium water level	Average minimum water level	Water level	Hydrograph
1	Achleiten	Achleiten	Donau	400 cm	2016-06-08 11:30	-	-	Unknown	ganglinien390
2	Passau Ilzstadt	Passau Ilzstadt	Donau	614 cm	2016-06-08 09:15	827 cm	418 cm	Normal	ganglinien294
3	Passau Donau	Passau Donau	Donau	616 cm	2016-06-08 09:30	832 cm	403 cm	Normal	ganglinien170
4	Vilshofen	Vilshofen	Donau	413 cm	2016-06-08 10:15	555 cm	299 cm	Normal	ganglinien494
5	Hofkirchen	Hofkirchen	Donau	397 cm	2016-06-08 09:00	557 cm	196 cm	Normal	ganglinien576
6	Deggendorf	Deggendorf	Donau	414 cm	2016-06-08 11:30	615 cm	192 cm	Normal	ganglinien344
7	Pfelling	Pfelling	Donau	497 cm	2016-06-08 09:15	697 cm	268 cm	Normal	ganglinien492
8	Straubing	Straubing	Donau	371 cm	2016-06-08 09:00	577 cm	123 cm	Normal	ganglinien236
9	Pfatter	Pfatter	Donau	413 cm	2016-06-08 10:00	601 cm	307 cm	Normal	ganglinien266
10	Schwabelweis	Schwabelweis	Donau	358 cm	2016-06-08 11:45	520 cm	283 cm	Normal	ganglinien428
11	Eiserne Brücke	Eiserne Brücke	Donau	320 cm	2016-06-08 09:15	501 cm	195 cm	Normal	ganglinien500
12	Niederwinzer	Niederwinzer	Donau	504 cm	2016-06-06 04:00	-	-	Unknown	ganglinien122
13	Oberndorf	Oberndorf	Donau	308 cm	2016-06-08 11:45	518 cm	157 cm	Normal	ganglinien186
14	Kelheimwinzer	Kelheimwinzer	Donau	353 cm	2016-06-08 11:45	516 cm	257 cm	Normal	ganglinien488
15	Ingolstadt Luitpoldstrasse	Ingolstadt Luitpoldstrasse	Donau	302 cm	2016-06-08 07:15	-	-	Unknown	ganglinien280
16	Schöna	Schöna	Elbe	173 cm	2016-06-08 11:30	641 cm	91 cm	Normal	ganglinien368
17	Pirna	Pirna	Elbe	199 cm	2016-06-08 11:15	614 cm	110 cm	Normal	ganglinien358
18	Dresden	Dresden	Elbe	167 cm	2016-06-08 11:30	574 cm	78 cm	Normal	ganglinien280
19	Meissen	Meissen	Elbe	224 cm	2016-06-08 11:15	637 cm	126 cm	Normal	ganglinien150
20	Riesa	Riesa	Elbe	239 cm	2016-06-08 11:30	635 cm	148 cm	Normal	ganglinien404
21	Muhlberg	Muhlberg	Elbe	262 cm	2016-06-08 11:15	684 cm	177 cm	Normal	ganglinien198
22	Torgau	Torgau	Elbe	167 cm	2016-06-08 11:30	623 cm	70 cm	Normal	ganglinien160
23	Pretzsch-Mauken	Pretzsch-Mauken	Elbe	165 cm	2016-06-08 11:15	584 cm	71 cm	Normal	ganglinien162
24	Elster	Elster	Elbe	165 cm	2016-06-08 11:15	514 cm	60 cm	Normal	ganglinien140
25	Wittenberg	Wittenberg	Elbe	233 cm	2016-06-08 11:45	543 cm	114 cm	Normal	ganglinien436

Import data Export data Map legend

Image source: https://www.7mmvalkyrie.com/load-data.html
https://www.gunwerks.com/store/featured-products/gunwerks-ammo
https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

Data Frame Format

- General form of data
  - Column = feature / attribute / variable
  - Row = object with feature / attribute values
- Example 1 :

There are 7 features

total_bill	tip	sex	smoker	day	time	size
16.99	1.01	Female	No	Sun	Dinner	2
10.34	1.66	Male	No	Sun	Dinner	3
21.01	3.5	Male	No	Sun	Dinner	3
23.68	3.31	Male	No	Sun	Dinner	2
24.59	3.61	Female	No	Sun	Dinner	4
25.29	4.71	Male	No	Sun	Dinner	4
8.77	2	Male	No	Sun	Dinner	2
26.88	3.12	Male	No	Sun	Dinner	4
15.04	1.96	Male	No	Sun	Dinner	2
14.78	3.23	Male	No	Sun	Dinner	2
10.27	1.71	Male	No	Sun	Dinner	2
35.26	5	Female	No	Sun	Dinner	4
15.42	1.57	Male	No	Sun	Dinner	2
18.43	3	Male	No	Sun	Dinner	4

Each row represents one bill with 7 feature values

Data Frame Format (cont.)

- Example 2

Data Dict : Meanings of features must be understood before data analysis

Pclass = Passenger class
SibSp = # of Sibling or Spouse
Parch = # of Parent or Children

Ticket = Ticket number
Cabin = Cabin number
Embarked = Embarking Part
(C = Cherbourg,
Q = Queenstown,
S = Southampton)

Passenger	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, Mrs. John Brad	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2.	7.925		S
4	1	1	Futrelle, Mrs. Jacques He	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr. James	male		0	0	330877	8.4583		Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta L	male	2	3	1	349909	21.075		S
9	1	3	Johnson, Mrs. Oscar W (E	female	27	0	2	347742	11.1333		S
10	1	2	Nasser, Mrs. Nicholas (A	female	14	1	0	237736	30.0708		C
11	1	3	Sandstrom, Miss. Margu	female	4	1	1	PP 9549	16.7	G6	S
12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103	S
13	0	3	Saundercock, Mr. William	male	20	0	0	A/5. 2151	8.05		S
14	0	3	Andersson, Mr. Anders J	male	39	1	5	347082	31.275		S
15	0	3	Vestrom, Miss. Hulda An	female	14	0	0	350406	7.8542		S
16	1	2	Hewlett, Mrs. (Mary D Ki	female	55	0	0	248706	16		S

Each row represents one passenger with 11 feature values

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Library Import

- All python libraries must be imported in order to be used
- Libraries can be renamed using as keyword

```
import pandas as pd
```


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Pandas Usage

- Data structures
  - 1D : Series
  - 2D: **Data Frame**
- I/O file manipulation
  - Example: read_X, to_X
- Basic information of data structures
  - Example: dtypes, columns, index, shape, len, head(), tail()
- Data selection
  - Example: indexing, slicing, Boolean indexing

Cartridge	Product Number	Bullet Make	Bullet Type	Bullet Weight Gr	COAL (Inches)
243 Winchester	M6000	Berger	VLD Hunting	90	2.775
6 Creedmoor		Berger	Hybrid Target	105	2.670
6XC	M6001	Berger	VLD Hunting	115	2.745
25-06 Rem	M6002	Berger	VLD Hunting	115	3.215
6.5 Creedmoor	M6004	Berger	Elite Hunter	140	2.865
6.5-284	M6006	Berger	VLD Hunting	140	3.085
264 Win Mag	M6007	Berger	VLD Hunting	140	3.325
6.5 PRC		Berger	Elite Hunter	140	2.970
280 Remington	M6012	Berger	VLD Hunting	140	3.330
280 Ackley	M6013	Berger	VLD Hunting	168	3.400
7mm RSAUM		Berger	VLD Hunting	168	2.800
7mm RSAUM	M6042	Berger	Hybrid Target	180	2.820
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7mm Rem Mag	M6015	Berger	VLD Hunting	168	3.420
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7mm Wby	M6017	Berger	VLD Hunting	168	3.340
7mm STW	M6018	Berger	VLD Hunting	180	3.670
7mm LRM	M6019	Berger	Hybrid Target	180	3.436
7mm RUM	M6020	Berger	VLD Hunting	180	3.665
28 Nosler	M6041	Berger	Hybrid Target	180	3.610
28 Nosler	M6042	Berger	Elite Hunter	195	3.630
308 Win	M6021	Berger	VLD Hunting	168	2.800
30-06	M6022	Berger	VLD Hunting	168	3.380
300 Win Mag	M6024	Berger	VLD Hunting	190	3.510
300 Wby	M6026	Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	210	3.650
300 RUM		Berger	VLD Hunting	190	3.650
300 RUM		Berger	VLD Hunting	190	3.650
30 Nosler	M6043	Berger	Hybrid Target	215	3.555
300 RUM	M6028	Berger	VLD Hunting	190	3.650
300 RUM	M6035	Berger	Hybrid Target	230	3.650
300 Norma	M6032	Berger	Hybrid Target	230	3.615
338 Norma	M6033	Berger	VLD Hunting	300	3.550
338 Lapua	M6034	Berger	VLD Hunting	300	3.600

Image source: https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

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Data Frame

- Basic information of data structures
  - dtypes, columns, index, , shape, len, head(), tail()
- Example

```
data = {'ID' : ['1001', '1002', '1003','1004'],
        'Name': ['Ann', 'Beth', 'Cathlyn', 'David'],
        'GPA': [3.51, 2.75, 3.04, 2.24]}
df = pd.DataFrame(data)
df
```

ID	Name	GPA
1001	Ann	3.51
1002	Beth	2.75
1003	Cathlyn	3.04
1004	David	2.24

	ID	Name	GPA
0	1001	Ann	3.51
1	1002	Beth	2.75
2	1003	Cathlyn	3.04
3	1004	David	2.24

```
df.dtypes
```

```
df.columns
```

```
df.index
```

```
df.shape
```

```
df.head(3)
```

```
len(df)
```

```
df.tail(3)
```


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Data Frame

- Data selection
  - Indexing, Slicing
- Example:

```
data = {'ID' : ['1001', '1002', '1003','1004'],
        'Name': ['Ann', 'Beth', 'Cathlyn', 'David'],
        'GPA': [3.51, 2.75, 3.04, 2.24]}
df = pd.DataFrame(data, columns=['ID','Name','GPA'])
df = df.set_index('ID')
df
```

ID	Name	GPA
1001	Ann	3.51
1002	Beth	2.75
1003	Cathlyn	3.04
1004	David	2.24

Index by label

```
df.loc['1002':'1003']
```

```
df.loc['1002':'1003', 'Name']
```

```
df.loc['Name']
df['Name']
```

```
df[['Name','GPA']]
```

Index by integer

```
df.iloc[2:3]
```

```
df.iloc[1002:1003, 'Name']
```

```
df.iloc[2:3, 0]
```

```
df.iloc[2:3, 'Name']
```


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Data Frame

- Data selection
  - Boolean Indexing
- Example:

```
data = {'ID' : ['1001', '1002', '1003','1004'],
        'Name': ['Ann', 'Beth', 'Cathlyn', 'David'],
        'GPA': [3.51, 2.75, 3.04, 2.24]}
df = pd.DataFrame(data)
df
```

	ID	Name	GPA
0	1001	Ann	3.51
1	1002	Beth	2.75
2	1003	Cathlyn	3.04
3	1004	David	2.24

```
df[df.ID == '1002']
```

```
df[(df.GPA > 2.5) & (df.GPA < 3.5)]
```

```
df[(df.ID == '1002') | (df.GPA > 3.0)]
```

```
df[(df.ID == '1002') | (df.GPA > 3.0)].Name
```


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Pandas Usage

- Data manipulation
  - Sort
  - Insert, Drop, Join, Replace
  - Missing data (null values) handling
- Mathematic functions
- Statistics functions
- And many more

Cartridge	Product Number	Bullet Make	Bullet Type	Bullet Weight Gr	COAL (Inches)
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7mm RSAUM	M6042	Berger	Hybrid Target	180	2.820
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Image source: https://www.subshell.com/en/sophora/documentation/add-ons/data-driven-journalism-with-maps/index.html

Conclusion

- Pandas is python library known for data manipulation and analysis
- In pandas, data frame structure are popularly-used
- In data science step, pandas is one commonly-used tool for exploratory data analysis